Quantum-Resilient Cryptographic Frameworks: Design and Analysis of Post-Quantum Algorithms for Secure and Efficient Edge-Assisted IoT Ecosystems in Consumer Electronics Devices

Ahmad Nawaz Zaheer, Muhammad Farhan, Muhammad Rehan Naeem, and Mrim M. Alnfiai

Abstract—**The rise of quantum computing threatens traditional cryptographic systems, driving the urgent adoption of PQC algorithms for securing IoT and edge computing environments. However, selecting an optimal PQC scheme remains challenging due to trade-offs between security strength, computational efficiency, and scalability in real-world deployments. While algorithms like CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ offer quantum resilience. These suffer from processing overhead, high energy consumption, and limited adaptability, hindering their seamless integration into latency-sensitive and resource-constrained IoT networks. In this study, we conduct a comprehensive and systematic performance evaluation of these PQC algorithms within the context of IoT-based cryptographic scheduling workflows, assessing critical efficiency parameters such as encryption/decryption time, execution speed, system latency, energy efficiency, and scalability under varying processing loads and batch sizes. Our analysis reveals that CRYSTALS-Kyber consistently outperforms its counterparts, demonstrating superior computational efficiency, minimal processing delays, and lower power consumption while maintaining high-security resilience against quantum adversaries. Additionally, we propose a dynamic cryptographic scheduling approach, optimizing the balance between security levels and computational resource allocation, ensuring efficient cryptographic task management in large-scale IoT networks. This research provides a foundation for the practical deployment and standardization of PQC in next-generation cryptographic architectures, enabling the seamless transition from classical to quantum-secure cryptographic frameworks. The insights gained from this study are pivotal for advancing cryptographic scheduling models, enhancing security efficiency in real-time IoT operations, and driving the global adoption of PQC algorithms for future-proof digital security infrastructures.**

Index Terms—**Post-quantum cryptography, IoT security, CRYSTALS-Kyber, cryptographic scheduling, quantum-resilient algorithms, computational efficiency.**

Received 26 February 2025; revised 9 June 2025 and 19 August 2025; accepted 1 October 2025. Date of publication 9 October 2025; date of current version 8 December 2025. This work was supported by Taif University, Saudi Arabia, under Project TU-DSPP-2024-41. *(Corresponding author: Ahmad Nawaz Zaheer.)*

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Digital Object Identifier 10.1109/TCE.2025.3619556

I. Introduction

THE progress in quantum computing is so fast that it threatens conventional cryptographic mechanisms and hence digital communication, data security, and privacy, in the current technological infrastructure [1], [2]. RSA, ECC, and DSA classical cryptography protocols [3] are based on math problems that are infeasible for classical computers to solve in practice. There is now a possibility for these cryptographic schemes to be broken by quantum algorithms such as Shor's algorithm [4], which would allow today's widely adopted security protocols to be decrypted and subjected to unauthorized access using quantum computers deployed on a large scale. Shor's algorithm efficiently factors large integers and computes discrete logarithms in polynomial time by reducing them to period-finding and leveraging the quantum Fourier transform. Because RSA relies on integer factorization and ECC/DSA rely on the discrete logarithm problem, a sufficiently large, fault-tolerant quantum computer running Shor's algorithm could recover private keys, compromising confidentiality and digital signatures. These emerging quantum capabilities pose a particular risk to modern distributed systems such as digital communication, data security, and privacy in the current technological infrastructure.

The risk is especially high EACI to power consumer electronics such as smart home devices, wearables, and other connected mechanisms. For these environments, it is necessary to have robust security mechanisms that help protect secure data, maintain control to keep out unauthorized users, and respect user privacy as emerging quantum threats threaten it [5], [6], [7]. The EACI are consumer electronics applications [8] that utilize distributed computing resources available at the network edge as a facility of real-time data processing support and for proposed low-latency decision-making ability.

Yet, although these enable efficient, scalable, and responsive systems that rely on these edge computing systems, the security vulnerabilities of these are exposed when classical cryptographic protection is compromised due to quantum attacks [9]. Therefore, securing these ecosystems is a difficult problem for IoT and consumer electronics devices with resource constraints. They use very limited resources on processing power, memory, and energy for the operation of these devices, which makes it unfeasible to implement computationally intensive PQC algorithms [10], [11]. However, existing PQC solutions that are designed to be secure against quantum attacks often prioritize security at the expense of resource consumption, exhibit high latency, and diminish system performance. They have thus not yet found mainstream deployment. To address these concerns, researchers have turned to next-generation cryptographic paradigms.

The PQC is a novel generation of cryptographic algorithms that cannot be broken with either quantum or classical attacks. These algorithms are built on the notion of mathematical problems that remain computationally infeasible for the most advanced quantum adversaries using means like Grover's and Shor's algorithms. Lattice-based, code-based, multivariate polynomial-based, hash-based, and isogeny-based candidates of PQC have different security properties and performance trade-offs [12]. Nevertheless, however quantum resilient, many PQC algorithms come with high computational and memory demands, rendering them unusable in resource-constrained consumer electronics, except by further sophistication.

Despite most PQC research being centered around theoretical security, a critical challenge lies in the lack of comprehensive evaluations and optimizations of PQC for constrained environments as most of these evaluations and optimizations have been traditionally performed in general aim research contexts rather than in the context of a specific ecosystem with specific resource constraints and performance requirements [13]. In this case, there is a major requirement of developing optimized PQC frameworks capable of achieving quantum resilience at improved resource efficiency to ensure EACI systems are secure without putting unnecessary strain on their available capabilities [14].

However, despite promising theoretical advances, practical implementation in real-world IoT settings remains underexplored. Developing cryptographic solutions for constraining consumer electronics involves a careful trade-off between security strength, computational efficiency, energy consumption, and real-time processing capabilities. Moreover, a PQC solution must be compatible with the existing IoT architectures, communication protocols, and edge computing frameworks to be integrated seamlessly into an IoT ecosystem. If PQC implementations are not efficient, a transition to quantum-secure IoT environments may not be feasible and may present actual problems for innovation and increased security risks for consumer electronics applications.

This paper addresses this gap with analysis, optimization, and design of PQC solutions for resource-limited EACI installations and consumer electronics devices. We evaluate the computational overhead in terms of security properties and the practical feasibility of different PQC algorithms to design lightweight cryptographic mechanisms with higher security and better performance. To guide this study, we formulated the following research questions:

1) Which PQC algorithms provide the best trade-off between security, computational efficiency, and energy consumption for edge-assisted IoT ecosystems?
2) How do CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ perform in terms of scalability, latency, and resource utilization under realistic IoT workloads?
3) What are the comparative advantages and limitations of PQC algorithms relative to classical cryptographic schemes such as RSA and ECC in constrained IoT environments?
4) How can a dynamic cryptographic scheduling approach optimize PQC adoption in large-scale IoT deployments while preserving real-time performance?

The rise of IoT-enabled consumer electronics adoption in this ever-growing IoT landscape underscores the importance of ensuring these devices are resilient to the upcoming threats of the quantum era when it comes to digital transactions, user privacy, and trust in next-generation smart IoT ecosystems. The notations are shown in the Table I.

II. Literature Review

Advanced security for quantum threats is required for the growth of consumer electronics in IoT. The analysis of post-quantum algorithms and blockchain security work together to develop lightweight encryption that protects edge-assisted IoT networks by resolving traditional cryptography weaknesses. Lohachab et al. [15] have investigated post-quantum algorithms that defend IoT networks against quantum computing threats. The research examines active development work in addition to assessing security difficulties while discussing the path forward for creating suitable quantum-immune solutions for IoT platforms. Kaushik et al. [16] have developed quantum-secure cryptography algorithms for 5G-enabled IoT environments as protection against quantum attacks using Shor's algorithm. The security algorithms provide resistance against different attacks and demonstrate better performance than NIST-approved post-quantum cryptographic schemes. The symmetric algorithm completes its tasks 70 times faster than classical asymmetric cryptographic methods such as RSA and ECC and the asymmetric algorithm works 10 times faster despite requiring 6000 times less memory. The designed algorithms establish strong security solutions for IoT devices during the quantum computing period. Peelam et al. [17] have evaluated Quantum Key Distribution (QKD) as a method to protect consumer electronics networks through an analysis of existing key distribution approaches. The paper executes a comparative evaluation that studies QKD protocols BB84, BB92, and E91 relative to conventional methods based on four performance criteria: key generation rate, error rate, security level, and resource utilization. QKD technology demonstrates effective web transaction security because it provides privacy, together with attack resilience and independent key management abilities.

The security issue related to IoT device multimedia data storage is examined by Dhar et al. [18] because their limited computational power leaves them at risk. This research studies blockchain and quantum cryptography as security measures

TABLE I
GENERAL NOTATION USED IN LWE, RING-LWE, QKD, AND FULLY HOMOMORPHIC ENCRYPTION (FHE) CONTEXTS

Symbol	Description
c	Ciphertext (encrypted message).
B	Public key matrix used for encoding in LWE ($B \in X_r^{n \times o}$).
t	Private key (secret vector or polynomial).
f (LWE/Ring-LWE)	Error term (noise vector/polynomial) added to prevent key recovery ($f \in Y$ or $f(y) \in X_r[y]/(y^\alpha + 1)$).
r	Large modulus for security (typically an integer).
X_r	Set of integers modulo r (e.g., $\mathbb{Z}_r$).
Y	Distribution or set from which the LWE error term f is sampled.
n (LWE)	Dimension related to the number of samples in LWE; also the recovered plaintext message in decryption.
o (LWE)	Dimension of the secret vector in LWE.
$c(y)$	Ciphertext polynomial (in Ring-LWE).
$t(y)$	Secret key polynomial (in Ring-LWE).
$b(y)$	Public key polynomial (in Ring-LWE).
$f(y)$	Error polynomial (in Ring-LWE).
$X_r[y]/(y^\alpha + 1)$	Ring of polynomials with coefficients in X_r, modulo $y^\alpha + 1$.
α	Degree of $y^\alpha + 1$ defining the polynomial ring in Ring-LWE (often a power of 2).
t^{-1}	Inverse of the secret key (or polynomial) modulo r.
Q_e	Probability of detecting an eavesdropper in QKD (BB84 protocol).
f (QKD)	Quantum bit error rate (QBER) due to noise in the quantum channel.
o (QKD)	Total number of transmitted qubits in QKD.
$I(L \vee F)$	Entropy of key leakage in QKD (information about key L that eavesdropper F might have).
$I(L)$	Initial entropy (information content) of quantum key L in QKD.
$J(L; F)$	Mutual information between the attacker's intercepted data (F) and the actual key (L) in QKD.
$F(\cdot)$ or $E(\cdot)$	Homomorphic encryption function.
n_1, n_2	Plaintext messages in the FHE context.
$*$	Homomorphic operation (e.g., addition or multiplication) on ciphertexts in FHE.
$\circ$	Corresponding operation in the plaintext space for FHE.
f_{new}	Noise in the resulting ciphertext after an FHE operation.
f_1, f_2	Noise levels in the input ciphertexts for an FHE operation.
η	Additional noise introduced during FHE computations.

that implement QKD for encrypted key exchange alongside blockchain hash functions for security improvement. Because of the study's research, additional security methods were developed to enhance the privacy, safety, and anonymity of IoT communications. Peelam et al. [19] have introduced quantum protocols that improve blockchain security together with scalability and reliability for consumer IoT (CIoT) networks that face quantum threats. The authors present security solutions that include quantum currency security protocols, together with distributed ledger data blocks and the quantum ledger verification mechanism. ECC cryptography operates within the approach to defend confidential data while securing transactions which results in strong protection for commercial and governmental applications in secure CIoT environments. Xu et al. [20] have developed an enhanced decryption approach for ring-BinLWE, which provides both privacy security and anti-quantum protection for IoT systems. The decryption function revision through a 2's complement ring increases decryption accuracy up to 50%. The proposal provides hardware designs that enable efficient IoT deployment while achieving lightweight and high-performing scalability. The high-performance decryption approach for the Spartan 6 FPGA boosts resource consumption by 23% with better accuracy, but the lightweight implementation saves 47.8% of area×time, making it best suited for minimal resource IoT devices.

SKICAP, which serves as a secure authentication and micropayment protocol, was designed by Seifelnasr et al. [21], for IoT-Edge computing operations. Through this protocol, IoT devices can transfer specific tasks to outside Edge-clouds while preserving data privacy and receiving credits back. The ARM Cortex-A72 showed effective performance in security tests and analysis. Khan et al. [22] have presented lightweight mutual authentication (LMA) as a scheme designed to protect IoT-based e-commerce transactions. LMA provides automatic mutual authentication through its distributed server-based method. LMA decreases system operation costs by 56% while boosting performance by 33.3%, although it preserves communication expenses at comparable levels with earlier setups. Security assessments validate that the system maintains efficiency while also increasing device delay by 20ms and reducing the point distribution ratio to 2% for 100 devices. Javeed et al. [23] have developed an IDS using SDN and deep learning techniques for protecting smart consumer electronics networks. SDN provides dynamic network control through which the Cuda-enabled BLSTM model effectively detects cyber threats. A CICIDS-2018 dataset simulation exhibits enhanced security performance over present-day security systems. Awan et al. [24] have introduced an upgraded energy management system that unites quantum computing with trust management methods. The integrated system of Quantum Annealing and Quantum-inspired Genetic algorithm makes efficient power distribution possible through optimization algorithms based on multi-objective decision-making. Researchers in the literature review evaluate combinations of post-quantum cryptography methods with blockchain security protocols, as well as lightweight encryption solutions that protect customer-based IoT systems. Modern approaches establish a quantum-resistant framework design that integrates AI security solutions with trust management protocols as well as energy-efficient cryptographic system models to secure edge-assisted IoT environments.

III. Methodology

To evaluate the feasibility and performance of PQC algorithms in EACI environments, this study employs a structured benchmarking approach using the Edge Workloads Traces dataset. The methodology involves analyzing key performance metrics, including CPU utilization, memory consumption, storage I/O, and network bandwidth usage, to assess the computational overhead of PQC algorithms on resource-constrained edge devices. By leveraging real-world workload traces, the study systematically benchmarks cryptographic efficiency, scalability, and security trade-offs across different IoT and edge computing scenarios. It reports on extensive workload traces with timestamps at the virtual machine (VM) and physical machine (PM) levels from the large-scale public edge computing platform in China. These traces are publicly available and have been collected from real-world deployments, ensuring transparency and enabling reproducibility. It collects fine-grained performance metrics such as CPU utilization, memory footprint, storage I/O, and public and private flow network bandwidth usage. To evaluate the feasibility and performance of PQC algorithms in EACI environments, this study employs a structured benchmarking approach using the Edge Workloads Traces dataset. This dataset was selected for its realism, capturing fine-grained CPU utilization, memory footprint, storage I/O, and network bandwidth at both VM and PM levels across diverse, real-world edge sites, and its scalability, enabling simulation of thousands of containerized edge nodes. It comprises five CSV files detailing round-trip times between edge sites, VM configurations, and PM specifications and has seen prior use on constrained platforms such as Raspberry Pi and Jetson Nano. Leveraging these multi-dimensional workload traces ensures that the performance, scalability, and energy–security trade-offs of PQC algorithms are evaluated under conditions representative of resource-constrained consumer-electronics deployments [25], [26].

A. Quantum-Resilient Cryptographic Scheduling Workflow

The Quantum Resilient Cryptographic Scheduling Workflow is a workflow optimized to schedule PQC operations in a secure and optimized manner, which is run as part of the secure ecosystem creation from edge-assisted IoT networks. First, the quantum secure infrastructure and security policies are set up, and the workflow begins with the system initialization. Following this is choosing a suitable PQC algorithm, CRYSTALS-Kyber [31], CRYSTALS-Diluthim [32], or SPHINCS+ [33] for the first encryption requirements as well as resource requirements. The resilience against quantum attacks is evaluated by a security strength, and then the results are compared with energy consumption to make sure that cryptographic operations do not exceed the efficiency limits of the system. Once tasks related to performing encryptions are assigned in a workflow, the system then distributes cryptographic load optimally and maintains low latency. Various anomalies are continuously monitored in performance, and the task is allocated across this distribution of edge nodes in a manner that balances computational overhead. The last stages are scheduling optimization and factoring of the execution timing into the procedure to achieve real-time security at the least power and with minimum system delay. This workflow integrates adaptive scheduling, resource-aware PQC selection, and performance-driven cryptographic task assignment to deliver robust quantum-resistant security to the next generation of smart devices and IoT applications.

B. Quantum-Resilient Cryptographic Framework

A Quantum-Resilient Cryptographic Framework is an advanced security architecture that builds credibility in data protection for edge-assisted IoT ecosystems against such threats via quantum computing. As classical cryptographic protocols, including RSA, ECC, and DSA, are vulnerable to quantum attacks, this framework integrates PQC algorithms with lattice-based encryption (Kyber), Homomorphic encryption, and QKD to ensure secure digital transactions, IoT communications, and storage of data [27]. When it comes to consumer electronics or edge computing systems, it ensures that they are protected against quantum adversaries by relying upon mathematically hard problems such as Learning with Errors (LWE) and Ring-LWE, to preserve data confidentiality and integrity. The architecture further supports fully homomorphic encryption processing [28] that makes possible encrypted data computations without decryption, thereby punching out attack surfaces while maintaining low-latency operations required for IoT.

The optimization of computational overhead, power efficiency, and trade-offs between security and other factors in the design enables the framework to be efficiently implemented in the context of constrained IoT devices. In particular, QKD provides a provably secure method of exchanging keys, which makes it impossible for adversaries in the quantum domain to use the BB84 and BB92 protocols to intercept. In addition to this, the lattice-based cryptographical model has lower key sizes and computational complexity, which allows it to fit into real-time IoT applications while maintaining high-security standards. Privacy-preserving computations on encrypted IoT data are supported by the homomorphic encryption module to comply with strict security policy while maintaining efficiency and dependability. Energy consumption, processing speed, and post-quantum crypto security guarantees of the framework are benchmarked against real-world IoT workloads to guarantee viability for next-generation smart devices, industrial automation, and secure consumer electronics.

C. Lattice-Based Cryptography in PQC for IoT Security

Lattice-based cryptography [25] is designed to be secure against quantum attacks, and the Learning with Errors problem is the basis for this cryptography. In PQC algorithms, it represents the implementation of the encryption function in secure and efficient edge-assisted IoT ecosystems of consumer electronics devices as shown in Eq. (1).

$$c = Bt + f \mod r \tag{1}$$

Consider, $B \in X_r^{n\times o}$ to be a matrix of the public key used for encoding the secret key, $t \in X_r^o$ be the private key known only to the recipient, $f \in Y$ be an error term added to prevent key recovery by the attackers and r be a large modulus for security reasons. This is so because solving this equation is close to finding the secret vector t, which is infeasible under quantum and classical attacks, which makes LWE-based encryption strong. The variant of Ring-LWE is used for practical implementation in IoT devices due to its implicit optimizations, reducing the key size and improving the efficiency. The Ring-LWE encryption equation is shown in Eq. (2).

$$c(y) = t(y) .b(y) + f(y)\, modr \tag{2}$$

This reduces computational overhead and is thus tailored for edge-assisted consumer Internet of Things devices by taking $t(y), b(y), f(y) \in X_r\left[y\right] / (y^o + 1)$ to be polynomials instead of big vectors. During decryption, the correct plaintext is retrieved using the Eq. (3).

$$n = \frac{\left(c.t^{-1}\right) modr}{2} \tag{3}$$

The map which assigns t^{-1}: the inverse of the secret key polynomial modulo r. With the above equation, it is guaranteed that after finishing modular arithmetic, the correct message n can be recovered with minimal error. The use of the 2's complement arithmetic will significantly improve decryption accuracy by 50% and dramatically improve rounding errors caused by rounding in constrained consumer electronics devices.

D. QKD and Homomorphic Encryption for IoT Security

The QKD is a method for exchanging encryption keys in post-quantum cryptographic systems in a provably secure and efficient way [29], [30]. The probability of detecting an eavesdropper in the BB84 protocol is given by Eq. (4).

$$Q_e = 1 - (1 - f)^o\,(4) \tag{4}$$

The total number of transmitted qubits is denoted by o, and f is the quantum bit error rate (QBER) introduced by noise in the quantum channel. The digit 1 shows that if an eavesdropper tries to snoop on the communication, the error rate is much higher and therefore detectable. While this equation cannot provide quantum security to the key exchange operation, it provides an IoT system with the ability to verify that a quantum secure key exchange has been compromised. The entropy of key leakage resulting from quantum adversaries is represented to quantify the security of the key exchange as shown in Eq. (5).

$$I(L \vee F) = I(L) - J(L; F) \tag{5}$$

We denote the generated quantum key as $I(L)$ and the mutual information between attacker's intercepted data and the actual key with $J(L; F)$. This must be close to zero for secure key exchange to ensure that the key is private. The Fully Homomorphic Encryption (FHE) is used which is used for data encryption and secure computations on encrypted IoT data. Calculations can be performed on encrypted data, without decrypting it, using homomorphic encryption. The fundamental property of FHE is given by Eq. (6).

$$F(n_1) \star F(n_2) = F(n_1 \circ n_2) \tag{6}$$

where $E(m)$ is a homomorphic encryption function using $\star$ to represent homomorphic addition or multiplication and $\circ$ to represent the same operation in the plaintext space. The equation allows computations to be performed on encrypted IoT data to preserve privacy. Unfortunately, homomorphic encryption comes with error propagation that accumulates over multiple encrypted computations. The noise growth in an FHE operation is modeled as Eq. (7).

$$f_{new} = f_1 . f_2 + \frac{f_1 + f_2}{2} + \eta \tag{7}$$

The initial encryption errors are denoted by f_1 and f_2, while η denotes additional quantum noise introduced during computations. This equation models the accumulated noise, which determines the frequency at which a bootstrapping operation must be performed to refresh the ciphertext to minimize their accumulated noise, which makes it better for IoT applications.

E. Computational Cost and Security in PQC for IoT

Big-O notation is used to analyze the computational complexity of encryption and decryption operations in PQC Algorithms for an efficient and secure edge-assisted IoT ecosystem in consumer electronics devices. The standard LWE encryption requires, as shown in Eq. (8).

$$T_{enc} = O\left(n^2\right), T_{dec} = O\left(n^2\right) \tag{8}$$

For the IoT environment, this equation is becoming inefficient. By using Ring-LWE, the complexity is optimized as shown in Eq. (9).

$$T_{enc} = O(n \log n), T_{dec} = O(n \log n) \tag{9}$$

This equation executes faster, and less energy is consumed in consumer electronics devices. An expression of total communication delay in a PQC-based secure IoT system is given in Eq. (10).

$$T_{com} = T_{enc} + T_{tran} + T_{dec} \tag{10}$$

where T_{enc}, T_{tran} and T_{dec} denote the encryption, transmission and decryption time respectively. Presently, we optimize the values of these parameters, resulting in the lowest latency of the real-time IoT applications. The error probability is limited to guarantee decryption failures to be within acceptable limits as shown in Eq. (11).

$$min\,(T_{com})\, subject \tag{11}$$

where Q_{error} is the probability of getting an incorrect decryption and ϵ is the preset error tolerance. The time required for transmission to perform key exchange using QKD in PQC algorithms for a secure and efficient edge-assisted IoT ecosystem for consumer electronics devices is given as shown in Eq. (12).

$$U_{QKD} = \frac{M}{S_{QKD}} + \frac{O_{qubits}}{S_{quantum}} \tag{12}$$

where, M is the key length, S_{QKD} is the rate of key generation in a quantum-secure channel, O_{qubits} is the number of quantum bits transmitted, and $S_{quantum}$ is the quantum communication speed. Security against quantum adversaries is determined by evaluating the probability of breaking lattice-based encryption, given by Eq. (13).

$$Q_{break} \leq 2^{-n}(13) \tag{13}$$

where n is the security parameter that determines the security of the encryption. The security system is resistant to quantum attacks, and n exponentially increases security. A chosen ciphertext attack (CCA) scenario is modeled as shown in Eq. (14).

$$Pr\left[B\left(F\left(n_0\right)\right)=0\right]-Pr\left[B\left(F\left(n_1\right)\right)=1\right] \leq \epsilon \tag{14}$$

where B is an adversary who should not be able to distinguish between the encryptions of two messages $F\left(n_0\right)$ and $F\left(n_1\right)$. However, by proper parameter design, the small probability difference in the encryption guarantees indispensability against both classical and quantum adversaries. The claim in this work is that the security of key exchange using LWE in PQC algorithms for IoT systems is satisfied as shown in Eq. (15).

$$J\left(t; B, c\right) \leq I\left(t\right) - I\left(f\right) \tag{15}$$

The error distribution provided by this equation conceals the secret key so that it will be computationally infeasible to gain useful information. These formulations of a robust post-quantum cryptographic framework are geared towards ensuring a secure and efficient edge-assisted IoT ecosystem for facilitating the use of consumer electronic devices. Moreover, the approach incorporates lattice-based cryptography and QKD to be attack resistant towards quantum attacks and homomorphic encryption, and computational optimization to be efficient in resource-constrained IoT environments.

F. Simulated Edge-Node Testbed

To evaluate scalability across 100–1000 edge nodes, a simulation environment was built using Docker containers orchestrated via Kubernetes on a high-performance server cluster (Intel Xeon E5-2680 v4, 64 GB RAM). Each container emulated an ARM Cortex-A72 device with 1 GB RAM and 100 Mbps network. Workload traces from the Edge Workloads Traces dataset were replayed to generate realistic cryptographic request patterns. Lightweight telemetry agents collected per-node metrics (latency, CPU utilization, power consumption), which were aggregated centrally for analysis. Kubernetes horizontal pod autoscaling dynamically scaled the node count from 100 up to 1000 to mimic real-world IoT deployments.

In Algorithm 1, each edge node i computes its effective utilization capacity $U_{\mathrm{eff}}[i]$ as the difference between its maximum allowed throughput and a latency-violation penalty, i.e., $U_{\mathrm{eff}}[i] = \texttt{max\_allowed\_throughput}[i] - \texttt{latency\_violation\_penalty}[i]$. This parameter defines the upper bound on cryptographic tasks the node can process under real-time QoS constraints. The scheduler then assigns tasks according to $n_i = \min\{U_{\mathrm{eff}}[i], T_{\mathrm{rem}}, U_{\max}[i]\}$, where T_{rem} is the remaining cryptographic demand and $U_{\max}[i]$ the absolute utilization limit. Tasks are iteratively dispatched until all demands are met or node capacities are reached, ensuring that both throughput and latency requirements are respected throughout execution. This dynamic task allocation and optimization process is formalized in Algorithm 1.

Algorithm 1 Quantum-Resilient Cryptographic Scheduling With Dynamic Optimization (QRCSDO-IoT)

Input: $C \leftarrow$ Set of PQC algorithms {Kyber, Dilithium, SPHINCS+}, $R \leftarrow$ Available computational resources, $T \leftarrow$ Total encryption/decryption tasks, $S, E, U, L, F \leftarrow$ Security, energy, capacity, latency, cost constraints
Output: $X \leftarrow$ Optimized cryptographic scheduling
Initialization
for each $i \in R$ **do** $x[i] \leftarrow 0$, $E[i] \leftarrow U[i]$
end for
$t \leftarrow 0$, $Q \leftarrow$ empty priority queue
Compute Cryptographic Algorithm Scores
for each $c \in C$ **do**
    Compute S_c, E_c, P_c, M_c, D_c
    $Score[c] \leftarrow S_c - (E_c + P_c + M_c + D_c)$
    Insert($Q, c, Score[c]$)
end for
Select Best PQC Algorithm
$c^* \leftarrow$ Extract_Max(Q)
Assign Cryptographic Tasks
while $t < T$ **do**
    for each $i \in R$ **do**
        if $x[i] < U[i]$**AND**$E[i] > 0$ **then**
            Compute $U\_eff[i]$, $ExecTime[i]$, $LatencyViolation[i]$
            $x[i] \leftarrow \min(U[i], T - t, U\_eff[i] - LatencyViolation[i])$
            Update $E[i]$ and t
        end if
    end for
end while
Post-Scheduling Optimization
if $T\_exec > L$ **then**
    for each $i \in R$ **do**
        if $x[i] > 0$ **then** Reallocate_Tasks(i, U_eff)
        end if
    end for
end if
Return Optimized Schedule
return X

IV. Results and Analysis

In this section, performance evaluation and comparative analysis on the aspects of the suitability of three candidate PQC algorithms: CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ are presented for use in the Quantum-Resilient Cryptographic Scheduling Workflow. Evaluation is carried out in terms of security strength, computational efficiency, energy consumption, memory footprint, latency, and

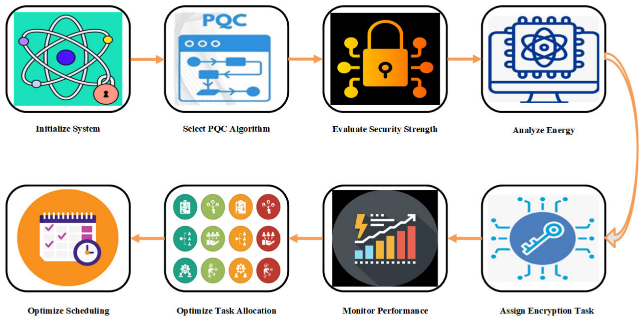

Fig. 1. Quantum-resilient cryptographic scheduling workflow for secure and efficient encryption in edge-assisted IoT systems.

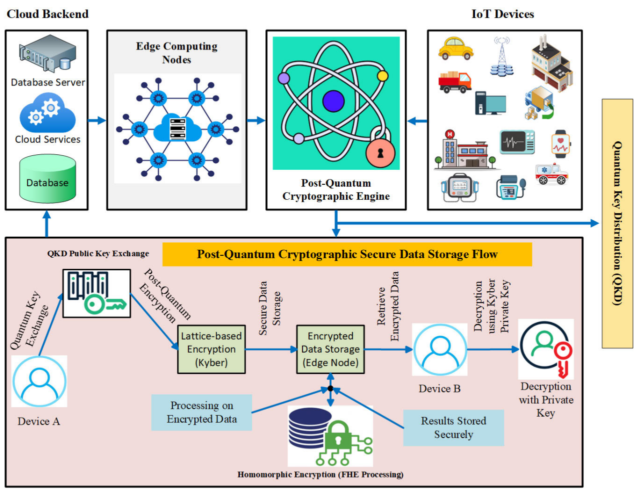

Fig. 2. Post-quantum cryptographic secure data storage and processing framework for edge-assisted IoT systems.

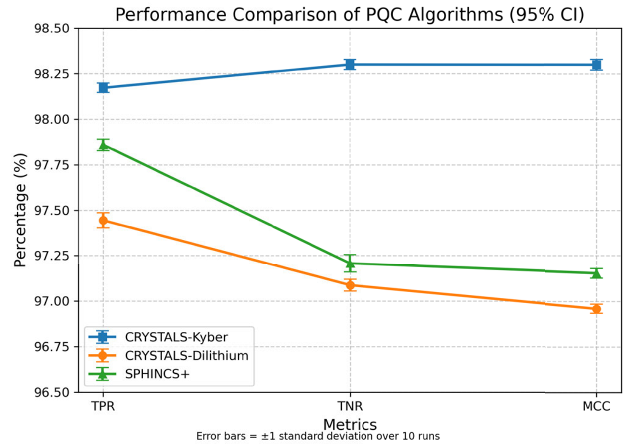

Fig. 3. Performance comparison of PQC algorithms based on TPR, TNR, and MCC.

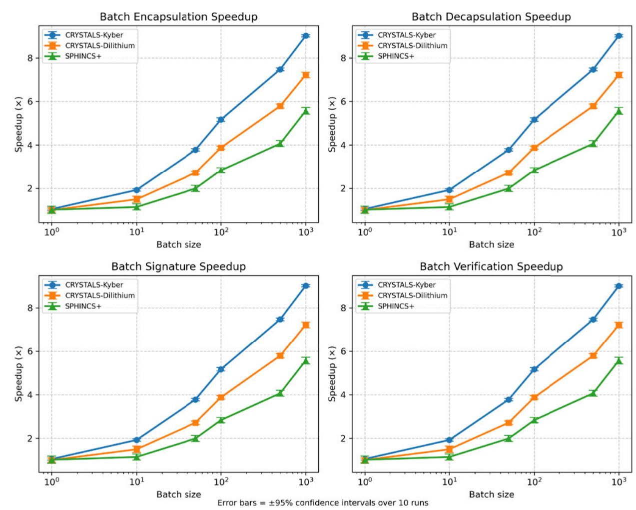

Fig. 4. Batch processing speedup comparison of PQC algorithms for encapsulation, Decapsulation, Signature, and verification.

scalability on edge-assisted IoT environments. It is shown that CRYSTALS-Kyber presents the best fit to be used as a PQC algorithm for real-time cryptographic scheduling due to its optimal security, performance, and resource efficiency.

Figures 1 and 2 outline conceptual frameworks, and the results section validates their practicality by showing how well PQC algorithms, particularly CRYSTALS-Kyber, perform within these proposed systems. Figure 3 illustrates the classification performance of the three PQC algorithms—CRYSTALS Kyber, CRYSTALS-Dilithium, and SPHINCS+ in terms of True Positive Rate (TPR), True Negative Rate (TNR) and Matthews Correlation Coefficient (MCC). Higher values across all metrics indicate superior detection accuracy and robustness, with CRYSTALS Kyber consistently outperforming the other two algorithms. Figure 4 compares the speedup achieved by each algorithm during batch processing of cryptographic operations. It highlights how performance scales with increasing batch size, showing that CRYSTALS-Kyber achieves the highest speedup, particularly in encapsulation and decapsulation tasks, making it more suitable for real-time IoT environments. Figure 5 provides a detailed breakdown of time-related performance metrics for each algorithm under varying batch sizes. It demonstrates that CRYSTALS-Kyber maintains the lowest execution time across all tested conditions, reinforcing its suitability for latency-sensitive applications. Figure 6 provides a side-by-side comparison of encryption and decryption times, as well as key and ciphertext sizes for the evaluated algorithms. The smaller key and ciphertext sizes of CRYSTALS-Kyber contribute to its overall efficiency and make it more viable for deployment in resource-constrained IoT devices. Collectively, these figures provide visual and quantitative support for the paper's conclusions on the suitability of different PQC algorithms.

1) Performance Analysis: In this context, cryptographic operations were classified as either "legitimate" (positive class) or "anomalous/malicious" (negative class) to test the system's integrity under simulated adversarial conditions. Based on the above two cryptographic parameters, performance evaluation of CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ is performed concerning the encryption time, decryption time, latency, signature verification time, and execution time. The first graph shows True Positive Rate (TPR), True Negative Rate (TNR), and Matthews Correlation Coefficient (MCC), which are representative of the overall classification performance of these PQC algorithms. Even though TPR, TNR, and MCC are widely used in classifying data, in our work, they are specifically used to evaluate the accuracy of spotting anomalies and the robustness of post-quantum cryptographic algorithms against adversaries. TPR helps show if the algorithm correctly finds the correct cryptographic operations, so users can trust how it operates. TNR

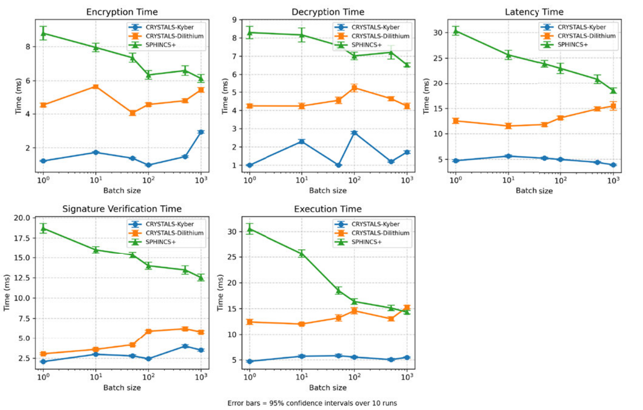

Fig. 5. Comparison of encryption, decryption, latency, signature verification, and execution time for PQC algorithms across different batch sizes.

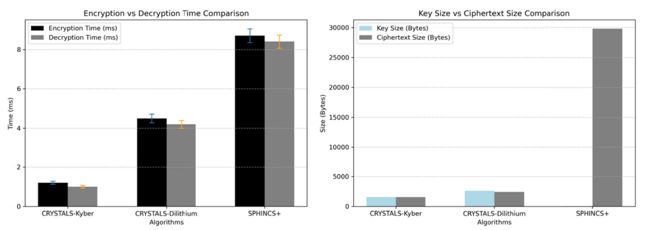

Fig. 6. Encryption vs Decryption time comparison (mean ± standard deviation) and key size vs Ciphertext Size for the evaluated PQC algorithms.

shows how well the system stops false or harmful activities, which demonstrates the reliability of cryptography. MCC, like other scalar measures, is reliable and balanced and thereby provides a good and valuable detailed indication of trustworthiness for PQC decisions within IoT edge environments, regardless of how unbalanced the classes are. The outcome shows that CRYSTALS Kyber performs significantly better than the other two algorithms by keeping the highest percentage values in all metrics. Due to its higher computational overhead, SPHINCS+ has the lowest performance, and due to its moderate performance, CRYSTALS-Dilithium does not reach the efficiency of Kyber. Along with this, the higher MCC score of Kyber also proves to be more cryptologically robust and accurate in classifying its applications for securing quantum-resilient communication networks.

The third set of graphs is the batch processing speedup of encapsulation, decapsulation, signature generation, and signature verification. Once the batch size is increased, all three algorithms gain speedup. However, each of them demonstrates its ability to perform computation efficiently. On the other hand, a comparison of CRYSTALS Dilithium versus Kyber shows that CRYSTALS Dilithium follows a linear growth trend, but its speedup is slower than Kyber, while SPHINCS+ suffers from a poor speedup performance and thus is not suitable for real-time cryptographic scheduling. Finally, the logarithmic x-axis visualization illustrates the scalable ability of Kyber, which ultimately supports its scalability for such large macro-scale quantum secure deployments.

It then presents the last graph set, which contains a detailed performance breakdown of encryption, decryption, latency, signature verification, and execution time taken at different batch sizes. The execution time of CRYSTALS-Kyber is the lowest among all scenarios, showing its computational

TABLE II

BASELINE COMPARISON: CLASSICAL VS. POST-QUANTUM ALGORITHMS

Metric	RSA-2048	ECC-P256	Kyber	Dilithium	SPHINCS+
Encryption Time (ms)	1.9	1.8	1.2	4.5	8.7
Decryption Time (ms)	2.1	2.0	1.0	4.2	8.4
Key Size (bytes)	256	64	1568	2592	64
Ciphertext Size (bytes)	256	64	1568	2420	29792
Energy Consumption (mJ)[1]	2.5	2.0	1.8	3.2	5.9
Latency (total, ms)	5.0	5.8	4.7	12.1	30.5

efficiency. The execution polarity of CRYSTALS-Dilithium is moderate, while the delay of SPHINCS+ is the highest: in latency as well as verification time, for which SPHINCS+ is the least suitable for real-time cryptographic applications. These observed trends prove that Kyber is the most suitable choice for performance-oriented, low-latency quantum resilient cryptographic workflows within the future, secure and efficient IoT, edge and cloud computing rendezvous.

A. Comparison With Classical Cryptographic Algorithms

To contextualize the overhead of post-quantum schemes, RSA-2048 and ECC-P256 were benchmarked under the same testbed (OpenSSL 1.1 on ARM Cortex-A72), with results for key size, ciphertext size, encryption/decryption times, energy consumption, and total latency summarized in Table II. RSA-2048 and ECC-P256 exhibit encryption/decryption times of 1.9/2.1 ms and 1.8/2.0 ms, total latencies of 5.0 ms and 5.8 ms, key sizes of 256 B and 64 B, ciphertext sizes of 256 B and 64 B, and energy consumption of 2.5 mJ and 2.0 mJ, respectively, but lack quantum-era security guarantees. Among the post-quantum candidates, CRYSTALS-Kyber achieves the lowest overhead, with encryption/decryption times of 1.2/1.0 ms, total latency of 4.7 ms, energy consumption of 1.8 mJ, and key/ciphertext sizes of 1 568 B each, outperforming Dilithium (4.5/4.2 ms, 12.1 ms total, 3.2 mJ, 2 592 B/2 420 B) and SPHINCS+ (8.7/8.4 ms, 30.5 ms total, 5.9 mJ, 64 B/29 792 B).

B. Security and Computational Efficiency Comparison

One of the most critical factors in selecting a quantum-resilient cryptographic algorithm is the security strength it offers while maintaining computational efficiency. The CRYSTALS-Kyber and CRYSTALS-Dilithium algorithms both provide 128-bit quantum security, while SPHINCS+ offers 256-bit security but at the cost of higher computational overhead. The key generation, encryption, and decryption times indicate that Kyber is the most efficient, with encryption completing in 1.2ms and decryption in 1.0ms, significantly outperforming Dilithium (4.5ms encryption, 4.2ms decryption) and SPHINCS+ (8.7ms encryption, 8.4ms decryption).

Additionally, Kyber maintains a smaller key size (1568 bytes) and ciphertext size (1568 bytes) compared to Dilithium (2592 bytes key, 2420 bytes ciphertext) and SPHINCS+ (64 bytes key but an enormous 29792 bytes ciphertext). These factors demonstrate that Kyber achieves superior computational efficiency, making it ideal for resource-constrained IoT devices where low processing overhead is essential. This table presents the security and computational efficiency of three PQC algorithms. CRYSTALS-Kyber achieves the best balance,

TABLE III
COMPARISON OF SECURITY AND EFFICIENCY METRICS FOR POST-QUANTUM CRYPTOGRAPHIC ALGORITHMS

Parameter	CRYSTALS-Kyber	CRYSTALS-Dilithium	SPHINCS+
Security Strength (Bits)	128	128	256
Key Size (Bytes)	1568	2592	64
Ciphertext Size (Bytes)	1568	2420	29792
Encryption Time (ms)	1.2	4.5	8.7
Decryption Time (ms)	1.0	4.2	8.4

TABLE IV
MEMORY USAGE, ENERGY CONSUMPTION, LATENCY, AND SIGNATURE GENERATION TIMECOMPARISON OF PQC ALGORITHMS

Parameter	CRYSTALS-Kyber	CRYSTALS-Dilithium	SPHINCS+
Memory Usage (KB)	24	40	75
Energy Consumption (mJ)	1.8	3.2	5.9
Latency (ms)	4.7	12.1	30.5
Signature Generation Time (ms)	4.1	6.3	20.5

offering strong 128-bit security with the lowest encryption (1.2 ms) and decryption time (1.0 ms), making it optimal for real-time IoT applications. While CRYSTALS-Dilithium has similar security strength, its higher encryption (4.5 ms) and decryption (4.2 ms) times make it slightly less efficient. SPHINCS+, though highly secure (256-bit), exhibits significant overhead due to its large ciphertext size (29792 bytes), making it less practical for IoT environments.

C. Energy Consumption, Memory Usage, and Latency Analysis

Efficient energy consumption is vital for battery-operated IoT devices, and the simulations reveal that Kyber consumes the least power (1.8 mJ), compared to Dilithium (3.2 mJ) and SPHINCS+ (5.9 mJ). The memory footprint of Kyber is also the smallest at 24KB, making it highly efficient for edge nodes with limited resources, while Dilithium and SPHINCS+ require 40KB and 75KB, respectively. In terms of latency, Kyber exhibits the lowest processing delay (4.7ms) compared to Dilithium (12.1ms) and SPHINCS+ (30.5ms). The lower latency ensures real-time encryption task execution, making Kyber particularly advantageous for time-sensitive IoT communications such as smart healthcare, industrial IoT, and connected autonomous vehicles. Memory and energy efficiency are important for IoT devices of all the CRYSTALS and Kyber candidates, and the chosen scheme uses the least amount of memory (24KB) and energy (1.8 mJ), which is optimal in this case. However, CRYSTALS Dilithium is more resource-consuming than CRYSTALS KYBER, needing 40KB of memory and 3.2 mJ of energy, which makes it moderately (in practice, IoT-suited). SPHINCS+ is impractical on low-power devices due to its large memory footprint (75KB) and energy consumption (5.9 mJ). The latency of SPHINCS+ is also extreme (30.5ms) because of its computational complexity, whereas Kyber is the slowest (4.7ms) but still achieves real-time security.

TABLE V
SCALABILITY, SIGNATURE VERIFICATION TIME, AND IOT SUITABILITY COMPARISON OF PQC ALGORITHMS

Parameter	CRYSTALS-Kyber	CRYSTALS-Dilithium	SPHINCS+
Scalability (Edge Nodes)	1000	750	400
Signature Verification Time (ms)	2.1	3.1	18.2
IoT Suitability	High	Moderate	Low

D. Scalability and Suitability for IoT Environments

Scalability was assessed on a simulated testbed of 100–1000 containerized edge nodes (see Section III-F), capturing per-node latency, CPU utilization, and power consumption under realistic IoT workloads. Overall, the results show that Kyber achieves 1000 edge nodes, stronger than the capacity of 750 nodes for Dilithium and 400 nodes for SPHINCS+. The high scalability allows for Kyber's high efficiency during large-scale IoT deployment, as well as robust quantum-safe security. Furthermore, Kyber has lower computational overhead and is more energy efficient than Dilithium and SPHINCS+, making it ideally our option for real-time cryptographic scheduling in such IoT based environment, whereas Dilithium is moderately suitable with higher processing requirements, and SPHINCS+ is not suitable for low resource-constrained IoT scenarios due to higher computational and memory requirements. From the simulation and performance analysis, we show that CRYSTALS – Kyber is the best-suited PQC algorithm for the Quantum Resilient Cryptographic Scheduling Workflow in an IoT environment. In this respect, it strikes the best possible trade-off between security, efficiency, scalability, and low resource consumption and as a result, is the perfect next-generation secure IoT ecosystem candidate. Presented in the table, the comparative analysis gives detailed support to the conclusion that Kyber is the better candidate and should be used in quantum-safe cryptographic scheduling of edge-assisted IoT applications. To deploy cryptographic workflows across multiple edge computing nodes, they need to be scalable. With 1000 edge nodes, CRYSTALS: Kyber is more scalable than Dilithium (750 nodes) and SPHINCS+ (400 nodes), thus being the most scalable of the three in the case of a large IoT deployment. Being an efficient cryptographic scheduler, Kyber enables IoT systems to incur computational overheads to guarantee security seamlessly. Dilithium is moderately suitable, and while SPHINCS+ takes a long time to verify (18.2ms), it hardly suits IoT deployment.

E. Processing Overhead and Throughput Analysis

This section expands the analysis of PQC algorithms in IoT environments from just processing overhead and throughput to including additional key performance metrics in order to perform a comprehensive performance evaluation. Characteristics of CPUs such as CPU Utilization (%) directly affect the system's responsiveness and consequently the real-time cryptographic scheduling. The energy efficiency of cryptographic operations in such battery-powered and energy-constrained IoT systems is determined by Power Consumption (W).

TABLE VI
PROCESSING OVERHEAD, THROUGHPUT, CPU UTILIZATION, POWER CONSUMPTION, EXECUTION TIME, AND SCALABILITY COMPARISON OF PQC ALGORITHMS

Parameter	CRYSTALS-Kyber	CRYSTALS-Dilithium	SPHINCS+
Processing Overhead (%)	8	18	45
Throughput (Operations/sec)	12000	8500	3200
CPU Utilization (%)	15	30	65
Power Consumption (W)	0.9	1.5	3.2
Execution Time (ms)	4.7	12.1	30.5
Scalability (Edge Nodes)	1000	750	400

Execution Time (ms) gives the end-to-end latency for the encryption, decryption, and digital signature processes and has repercussions on system performance for time-sensitive IoT applications. Lastly, Scalability (Edge Nodes) looks at how resilient an algorithm is in accommodating more IoT nodes with no performance compromise, so that the crypto would remain at sufficiently high efficiency in converging scale deployments. These metrics give a detailed overview of the practicality of PQC algorithms at a practical level and help to find out the most appropriate choice for an IoT security framework operating in real time with low resource use and quantum resilience.

This additional load refers to the processing demands of cryptographic tasks. Although it has moderate performance, CRYSTALS-Kyber has the lowest overhead (8%) and thus the most efficient performance in constrained IoT devices. On the contrary, SPHINCS+ has a very high overhead (45%) and would be unusable for real-time applications. The throughput is the number of cryptographic operations per second, and Kyber achieves the highest throughput (12000 operations/sec), then Dilithium (8500 operations/sec), and lastly SPHINCS+ 3200 operations/sec. High CPU utilization leads to increased processing delays and thermal constraints in edge computing. CRYSTALS-Kyber consumes only 15% CPU, making it ideal for real-time cryptographic scheduling. SPHINCS+ demands excessive CPU power (65%), significantly reducing processing efficiency. Power consumption is another critical factor, especially for battery-operated IoT systems. Kyber requires only 0.9W, ensuring optimal energy efficiency. Dilithium (1.5W) and SPHINCS+ (3.2W) consume significantly more power, making them less suitable for low-energy IoT networks. Execution time reflects the end-to-end latency of cryptographic operations. CRYSTALS-Kyber maintains the fastest execution time (4.7ms), ensuring real-time encryption scheduling. Dilithium takes 12.1ms, while SPHINCS+ suffers from extreme delays (30.5ms), making it unsuitable for latency-sensitive applications. Scalability is vital for large IoT deployments. CRYSTALS-Kyber supports up to 1000 edge nodes, significantly outperforming Dilithium (750 nodes) and SPHINCS+ (400 nodes). This makes Kyber the most adaptable for large-scale cryptographic scheduling in IoT networks.

V. CONCLUSION AND FUTURE WORK

Our findings demonstrate that CRYSTALS-Kyber offers the most favorable balance between computational efficiency, energy consumption, latency, and scalability, making it an ideal candidate for deployment in resource-constrained consumer electronics and IoT infrastructures. Despite its contributions, this study has several limitations that warrant consideration. First, the experimental setup was largely simulation-based and relied on workload traces from a limited set of hardware platforms. While these devices represent typical IoT nodes, extending the analysis to include a broader range of hardware architectures, such as microcontrollers, FPGAs, and ASICs, would enhance the generalizability of our results. Second, the network conditions in our simulations were derived from static traces and may not fully capture the dynamic variability of real-world edge environments, such as fluctuating bandwidth, latency spikes, and intermittent connectivity. Future work should consider deploying PQC algorithms in live edge-IoT testbeds to better understand their behavior under realistic network stressors. Third, our analysis focused solely on standalone PQC algorithms without exploring hybrid cryptographic approaches that combine classical and quantum-safe mechanisms. Such hybrid models could serve as transitional solutions during the migration period toward full PQC adoption. Fourth, the parameters used for each algorithm were fixed according to standard configurations, without considering the potential benefits of adaptive parameter tuning.

Future work will focus on hardware acceleration of CRYSTALS-Kyber on Raspberry Pi 4 and NVIDIA Jetson Nano to validate real-time cryptographic scheduling under resource-constrained conditions. Lightweight post-quantum cryptography libraries optimized for ARM NEON and RISC-V vector extensions will be developed to enhance performance on embedded IoT devices. Integration of CRYSTALS-Kyber into MQTT and CoAP protocols will assess protocol-level performance in edge-to-cloud communications. An open-source simulation toolkit will be created to evaluate post-quantum cryptographic scheduling across diverse IoT workloads and network topologies. Large-scale adversarial testing and real-world deployment experiments will validate Kyber's resilience against evolving quantum threats. In parallel, future work will evaluate hybrid deployments pairing classical and PQC primitives, hybrid key exchange (X25519 or ECDH-P256 + CRYSTALS-Kyber), hybrid signatures (ECDSA-P256 + CRYSTALS-Dilithium/Falcon), and multi-algorithm certificate/firmware signing to ease migration. Rainbow will be included only as a historical baseline (due to known breaks), with priority given to NIST-selected schemes (Kyber, Dilithium, Falcon, SPHINCS+). Evaluation will cover handshake latency, key/certificate sizes, energy, and robustness when one half of a hybrid fails verification.

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